



Teaching History With Custom-Built Board Games

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Abstract

Purpose. This article investigated the potential of a **custom-built board game** as a means to teach **historical empathy**, improve class **participation**, and to improve **student understanding** of the limitations of archival collections.

Method. Participant behavior as well as verbal and written feedback were collected during a pilot study of POLICING THE SOUND. A total of 88 **undergraduate** and **graduate** participants from History and Indigenous Studies took part during the pilot study.

Results. Student participation and understanding of historical context improved during the game. While graduate and undergraduate students showed similarity in their **enjoyment** of the game and their belief that it made historical arguments, the **curricular differences** in graduate and undergraduate programs influenced how each group approached the game.

Conclusion. Participant feedback, facilitator observation, and external observation indicate that groups of players can resolve confusion more efficiently than individual players can, time constrained decision-making may help maintain student engagement, an inability to win does not necessarily cause disengagement in short educational games, and that a structured debrief is important in achieving educational goals even in custom-built games.

Keywords

borders, custom-built games, educational goals, empathy, engagement, games and learning, graduate, historical simulations, history, history games, history simulation, history teaching, participation, undergraduate

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Educators have been using games for hundreds of years to teach everything from history and military tactics to ethics and morality (Garris, Ahlers, & Driskell, 2002; Nicholson, 2013; Parker Brothers, 1895; Smith, 2013; Wilson et al., 2009). At their core, games are non-obligatory, unproductive, and pleasure-providing activities that are bound by rules and contain meaningful choice. Simulations share many aspects with games, but attempt to simplify dynamic systems that exist in the real world with a much greater degree of veracity (McCall, 2012, 2016). While scholars continue to disagree about the exact definition of what constitutes a game, simulation, play, or the ludic, most agree that games and simulations are more than just amusing pastimes (Cailliois, 1961; Corbeil, 1999; Huizinga, 1970; McCall, 2016; O'Neil, Wainess, & Baker, 2005; Rockwell, 2002; Vowinckel, 2009). Since at least the end of the nineteenth century, toy manufacturers, educators, and doctors have recognized the significant value of games as forms of instruction (Hutchinson, 1908; McLoughlin Brothers, 1897; "Toys of War Wanted," 1899).

In a modern educational setting, games and simulations provide controlled, immersive, and interactive environments that shift as participants make decisions. In medicine and aviation, for example, the controlled nature of simulations has allowed doctors, nurses, and pilots to learn the practical motions of their trades without having to risk human lives or expensive equipment in the process (Jöud et al., 2010; Wilson et al., 2009). In history classrooms, the interactive and dynamic nature of games has allowed students to experience historical settings and systems rather than simply reading about them. While students can visit the ruins of a medieval city, they cannot physically visit previous eras to see cultural processes in action or to experience the shifting relationship between cause, context, and effect. Games and simulations allow students to interact with past environments, albeit limited recreations of them, in ways not otherwise possible (Clyde, Hopkins, & Wilkinson, 2012; Corbeil & Laveault, 2011; Kee et al., 2009; McCall, 2016; Morgan, 2013; Vowinckel, 2009).

Over the past 20 years, hundreds of scholars have studied the application of computer games to learning environments. Ricci, Salas, and Cannon-Bowers (1996) have concluded that computer-based games have improved knowledge retention over reading and test based approaches. Randel, Morris, Wetzel, and Whitehill's (1992) macro-analysis of 68 studies on games (both digital and non-digital) within math and the sciences, social sciences, and languages concluded that games provide situational benefits depending on the type of learning undertaken. More recently, scholars have investigated the impact that competitiveness (DeLeeuw & Mayer, 2011), motivation and instructional supports (Fiorella & Mayer, 2012; O'Neil et al., 2005), research design (All, Castellar, & Looy, 2016), game form (Rockwell et al., 2013), immersion and empathy (Bachen et al., 2016), and the type of change desired have on the efficacy and evaluation of game-based learning activities (Wilson et al., 2009).

Within the discipline of history itself, Fisher (2011) has argued that commercial off-the-shelf products like *Medal of Honor* and *Call of Duty* capture student interest in ways that traditional presentations of history do not. The analysis of games like *The Age of Empires: Conquerors* by Yu, Yu, Fan, and Wang's (2013) and *Diplomacy* by Corbeil (1999) as well as the *Assassin's Creed*, *Call of Duty*, *Total War*, *Mission U.S.*,

and *Civilization* franchises by McCall (2016) and Wainwright (2014) suggest that games aid in student understandings of why the past happened by allowing them to engage in counterfactual reasoning. While studies of commercial off-the-shelf games dominate, studies on the possible efficacy of custom-built games or heavily modified commercial off-the-shelf games are increasing in number. Studies by Kee (2011), Smith (2013), and Admiraal et al. (2011) suggest that custom-built or modified games show promise in teaching a variety of historical epistemologies, increasing student engagement, and overlaying historic lessons on top of contemporary landscapes. The rising popularity of board games in the past decade has renewed scholarly interest in using board games, rather than simply computer games, in educational environments (Sardone & Devlin-Scherer, 2016).

This article contributes to the scholarship on games and simulations by expanding the literature around the use of non-digital custom-built games in educational settings and by incorporating graduate students, rather than simply undergraduate students, in the analysis. Non-digital games share broad similarities with their digital and commercial off-the-shelf counterparts but offer a unique set of logistical challenges to teachers. This article examines the pilot run of *POLICING THE SOUND*, a custom-built board game, to 88 participants across five undergraduate and graduate classes at the University of Saskatchewan in Canada. Based on user-feedback and external observation, this paper argues that custom-built board games increase student participation and offer a means to not only teach facts (names and dates) and historical systems, but also to teach complex ideas and approaches, such as research methodologies and historical empathy. More broadly, the initial results from *POLICING THE SOUND* indicate that groups of players can resolve confusion more efficiently than individuals can, time constrained decision-making may help maintain student engagement, an inability to win does not necessarily cause disengagement in short educational games, and that a structured debrief is important in achieving educational goals even in custom-built games.

The Challenge of Games in the Classroom

Games, whether digital or tabletop, create unique challenges to the teaching of history. As Rosemary Garris, Robert Ahlers, and James E. Driskell have argued, the term *instructional game* is [in one sense] an oxymoron. Game play is voluntary, nonproductive, and separate from the real world. Instruction or training [by contrast] is typically nonvoluntary, undertaken to achieve certain learning outcomes, and related to life or work skills” (Garris et al., 2002, p. 459). Moreover, the principles of good game design, which includes making meaningful decisions, have an uncomfortable relationship with the historical profession, which emphasizes the need to study what happened and why and often shies away from counterfactuals (Robison, 2013). In practice, many historically themed commercial off-the-shelf games or franchises, such as *Call of Duty* and *Medal of Honor*, emulate movies far more than they do actual historical events. The *Civilization*, *Age of Empires*, and *Command and Conquer* franchises add excitement to history by applying ahistorical timelines, counterfactual narratives, or

supernatural overtones to historical settings (Campbell, 2008; McCall, 2012; Reading & Harvey, 2008).

Even when commercial game designers render the general setting of the game accurately, the game mechanics they choose can send misleading messages. As James Campbell has argued, World War II first-person shooters provide a skewed sense of conflict in which “speed and accuracy with infallibly reliable weapons make the difference between victory and defeat” (Campbell, 2008, p. 197). In reality, the accuracy of a weapon played far less of a difference to the outcome of combat in World War II than the number of bullets a soldier fired. Historically themed games often balance the quest for historic accuracy with the necessity for meaningful choice by allowing for small-scale personal victories while leaving the broader historical events unchanged. This approach, while clever, refocuses discussions away from structural factors that shaped history and onto personal narratives creating a skewed sense of importance (Campbell, 2008; Fullerton, 2008; Kim, Park, & Baek, 2009). Finally, even when games create nuanced historical environments, students do not always engage with them the way educators anticipate. Students skip over in-game text and videos in order to get to their gameplay goals faster limiting the kinds of educational goals that teachers can achieve (Admiraal et al., 2011).

Digital games also create a wide range of logistical hurdles during implementation. As Eric Morgan (2013), Devlin-Scherer and Sardone (2010), Watson et al. (2011), McMichael (2007), and Charsky and Mims (2008) have highlighted, computer games create challenges for teachers because they require access to a large bank of computers, licensing agreements, and personnel to troubleshoot inevitable technical difficulties. These games can also take hours to complete requiring teachers to either dedicate large amounts of class time to them or cut the student’s interaction with the game short before they have had much more than a taste of the game’s progression.

Tabletop games, by contrast, do not require the same kinds of technical supports in order to implement them but bring a whole host of challenges of their own. Historically themed board games, such as *New Amsterdam* (Allers, 2012), *Lewis & Clark* (Chaboussit, 2014), *Dakota* (Cioni, 2010), *Kolejka* (Madaj, 2011), and *Freedom: The Underground Railroad* (Mayer, 2012), for example, require students to handle all of the in-game bookkeeping and rule enforcement (something computer games do automatically). While this can increase student understanding of the game and perhaps the learning outcomes associated with it, detailed rule explanations add a fair amount of additional set-up time (Drake & Sung, 2011). Tabletop games similarly do not allow participants to save the game state easily requiring that facilitators either reserve physical space in their classrooms to store the in-progress games or limit the games played to only those that can be played within a single class. While commercial off-the-shelf games hold promise in the classroom, the specific nature of the market they appeal to limits their use in many university-level classrooms. While there are dozens of commercially available games based on military conquest or civilization building, there are far fewer games related to other fields of history, including social and cultural history, that are frequently taught at a university level.

Policing the Sound

POLICING THE SOUND is a game-based learning activity derived from archival research that attempts to challenge players' assumptions about criminality, government power, cultural production, and source base by allowing players to make decisions within a constrained historical environment. It plays 8-50 players, in 50-120 minutes including debrief, using custom-made cards and components. POLICING THE SOUND was created out of frustration with traditional teaching methods and existing commercial games. My past attempts to teach the history of international smuggling and federal border control through lectures and primary/secondary readings succeeded in teaching students about the content, but did not stop them from drawing stark lines between lawmakers and lawbreakers. Many of the students continued to believe that if they were placed in similar circumstances, they would not break the law. This made it difficult for them to empathize with the historic realities of borderland life and to understand the often uneasy and sometimes unclear divide that nineteenth century communities drew between smugglers, immigrants, merchants, and law enforcement officers (Lew-Williams, 2011).

Unsurprisingly, no commercially available game existed on smuggling along the BC-Washington border in the nineteenth century. Moreover, the kinds of logistical challenges - variable class duration, changing numbers of students, and limited access to equipment - that are common in university and high school classrooms would have impeded the implementation of an off-the-shelf product even if one had existed (Devlin-Scherer & Sardone, 2010; McMichael, 2007; Raphael et al 2010; Watson et al., 2011). POLICING THE SOUND aimed to address both the logistical challenges of a variable classroom and the shortcomings of traditional teaching methods.

The game was developed by taking an existing game mechanic (bluffing/risk assessment) from a commercially available game (Sheriff of Nottingham) and building a completely new asymmetrical game around it based off archival records. Solo and small group playtesting sessions honed the game until it could be explained and played in a very short time frame (Halaban & Zatz, 2014). Game components were designed in adobe illustrator and fabricated by hand using cardstock and illustration board. Later versions of the game utilized cards that can be printed cheaply from websites like Printer Studio, CowCow, and ArtsCow to increase the polish of the game components.

In POLICING THE SOUND, players are divided into nine teams. Eight teams represent West Coast communities (New Westminster, Point Roberts, Port Townsend, Neah Bay, Bellingham Bay/Lummi Island, Victoria, Colville, and Metlahatla/Haida Gwaii) that archival research indicated smuggled various goods across the border. The final team plays as the customs inspectors whose task it is to capture smugglers (Figure 1).

At the start of the game, the facilitator shows the participants a map of where their communities are located and provides them with the rules and a brief backstory. POLICING THE SOUND is played over five rounds starting in 1879 and progressing in three year intervals until 1892. Every round each community draws cards (ranging from 2 to 5) from a deck of products that are specific to each community. Each team

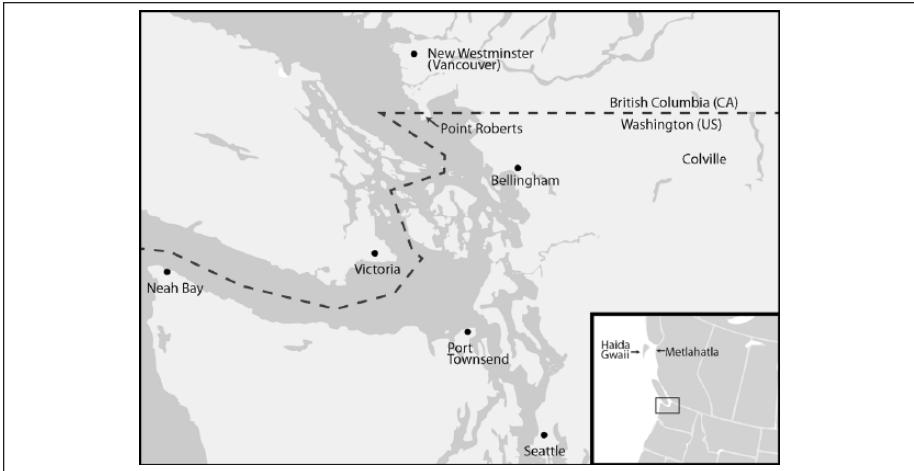


Figure 1. Map of Puget Sound.

then simultaneously plays cards facedown onto their player board choosing to either smuggle their cargo, transport it legally, or a combination of both. Smaller communities can only play two cards per turn while the largest ones can play five (Figure 2). At the end of each round, the customs inspector chooses to investigate one or, after 1888, two communities. Communities caught smuggling receive a small fine and do not gain proceeds for their smuggled cargo. Any smuggled cargo caught by the customs inspectors adds to their score. This incentivizes the customs inspectors to target large port cities that play more cards each round and communities that have access to the kinds of goods that have high profit margins when smuggled. After the customs inspectors have acted, all other communities then reveal their income, but do not indicate whether they smuggled. The facilitator tracks the total wealth of each community and makes the running total available to all teams.

The value of the cargo in *POLICING THE SOUND* changes over time to reflect the historical progression outlined by the Customs and Immigration records held at the National Archive in Seattle (United States Customs Service. (1886-1912); United States. U.S. District Courts Western District Washington.). This transition in payouts, while abstracted into a game environment, aims to simulate the borderland environment in the nineteenth century in a rough sense. At the end of five rounds, the team that accumulated the largest amount of wealth wins.

After the conclusion of the game, the facilitator ran a debriefing session that lasted approximately forty minutes. This debriefing took the form of a series of semi-structured questions (Appendix A), which students responded to verbally. During the debriefing, the facilitator posed the questions to the entire class at once, rather than having the smaller groups discuss them individually. Large discussion questions were interspersed with informal surveys of the participants (“with a show of hands how



Figure 2. Game Components for POLICING THE SOUND.

many of you...” to keep the full class engaged and targeted prompts (“for the team playing the customs agents...”) to tease out larger patterns of behavior. This style of debriefing worked very well in this context because of the asymmetrical nature of the game. Having students discuss the game as a large group made the in-game differences in experiences immediately evident and allowed each group to share the strategies that they employed. This debriefing strategy was created in tandem with the game design process itself. In this sense, the debriefing questions served as a means to bring student’s attention to the structural systems the game simulated well, as well as to highlight the portions of history that the game failed to represent. As mechanics (bribery etc) were cut from the game during playtesting to streamline the experience, debriefing questions were created in their place to ensure that key learning objectives were still achieved. In this context, the debriefing served as a foundational part of the learning experience by allowing the students to understand the attributes of each of the asymmetrical roles and how they interacted without having to play the game multiple times to experience those perspectives firsthand. While learning therefore occurred in the game environment, the debriefing allowed students to increase the breadth of their learning by combining experiences across teams, and increasing the depth of learning by targeting student attention to key areas of analysis.

Lessons for Implementing Custom Built Games in Classroom Environments

At the end of each session of POLICING THE SOUND, participants recorded anonymous comments about what they liked or disliked about the game and ranked the game on a scale of one to five based on three questions:

1. “Was playing “POLICING THE SOUND” enjoyable?” (5 - Very Enjoyable, 3 Moderately, 1 not at all)
2. “Did the game make historical arguments?” (5 Absolutely, 3 some, 1 not at all)
3. Depending on the iteration of the game:
 - a. “Could the arguments have been made better through readings” (5 Yes, 3 Maybe, 1 Not at all)
 - b. “Would you have understood the history of smuggling and its complexity better if you had spent the class listening to me lecture instead?” (5 No, 3 Similar, 1 Yes)

Participants (n=88) filled out anonymous feedback forms rating the game on average (mean) as very enjoyable (4.7 out of 5) and believed it made a clear historical argument (4.5 out of 5). The vast majority of participant rated the game at least a 4 out of 5 in terms of enjoyment and believed POLICING THE SOUND made clear historical arguments (4 or higher). The anonymous feedback gathered during the sessions of POLICING THE SOUND reinforces the broader conclusions that scholars of commercial off-the-shelf games have made about the value of games for creating an immersive and enjoyable experience in the classroom (McCall, 2016). External observation, coupled with the anonymous feedback from undergraduate and graduate participants during the pilot run of POLICING THE SOUND, suggests seven potential lessons to other teachers and designers interested in utilizing custom-built board games in classroom environments.

Lessons 1: Participant Interest in Teaching Impacts the Reception of Game-Based Learning Strategies

While a large number of studies have been conducted on the use of educational games in K-12, undergraduate, and professional schools such as law (Bird, 2002; Bridge, 2013), nursing (Chen et al., 2015), engineering (Bodnar et al., 2016; Ebner & Holzinger, 2007), business (Faria, 2001), and medicine (Fukuchi et al., 2000; McCoy et al., 2016; Robertson & Tannahill, 2009), there remains a persistent gap in the analysis of the use of games in MA and PhD programs. The post-secondary studies that have been created have tended to focus on capturing participant responses at either the undergraduate or graduate/professional level but not both (Eck, 2006; Wu & Hwang, 2012). This has made it difficult to determine how to translate games across degree programs in universities, where teachers commonly teach both graduate and undergraduate courses.

The pilot run of POLICING THE SOUND captured both undergraduate and graduate responses to the same game. Over the initial pilot run, five classes participated in

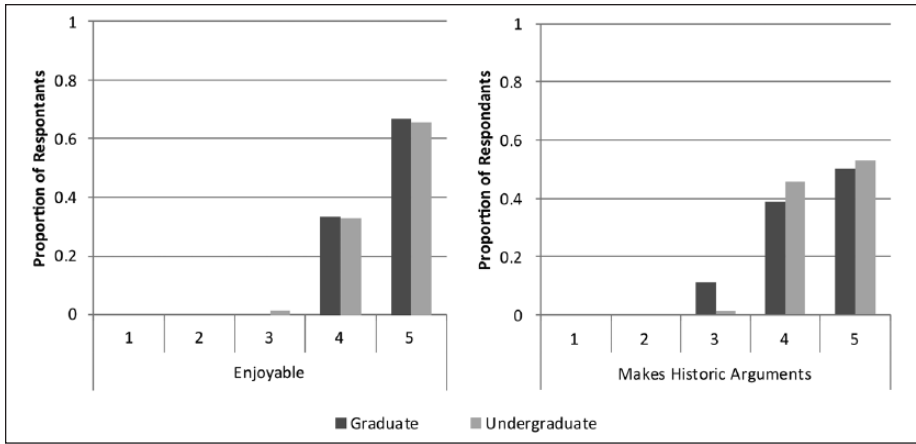


Figure 3. Undergraduate and Graduate Student Reflections.

playthroughs of *POLICING THE SOUND*. At the undergraduate level, 61 students from two separate second-year American History survey courses participated as did 9 students from a fourth-year history seminar on Aboriginal History. At the graduate level, 18 M.A. students (from Indigenous Studies and History programs) from two separate iterations of a Methods in Historical Research course took part.

Participant reflections and external observation indicate that both graduate and undergraduate students had roughly the same experiences while playing *POLICING THE SOUND* in terms of their own enjoyment and their perception about the possibility of games making historical arguments (Figure 3). Student comments at both levels noted an enjoyment with seeing history interactively “in a physical form” and valued the game for linking the history “so well with what really happens.”

Graduate students ($n=7$) responded a bit more critically than undergraduate students ($n=28$) to questions about how well they believed the game taught history in comparison to traditional teaching methods (Figure 4). The primary difference in the ways the audiences engaged with *POLICING THE SOUND*, however, emerged in the verbal and written comments.

Although the game was used at both a graduate and undergraduate level for the same general purpose (fostering historic empathy and emphasizing limitations in archives), graduate student feedback often focused on how they might use a game like this to teach others rather than on what they themselves learned from the experience. This differed substantially from undergraduate student comments, which focused on the game’s content and on their own learning. Graduate student concerns and experiences as teaching assistants (TAs) appears to account for this difference. This result emphasizes that prospective game designers should take into consideration the surrounding tasks students are engaged with or thinking about (in this case TAing) as it can have a noticeable impact on how students approach game-based learning activities.

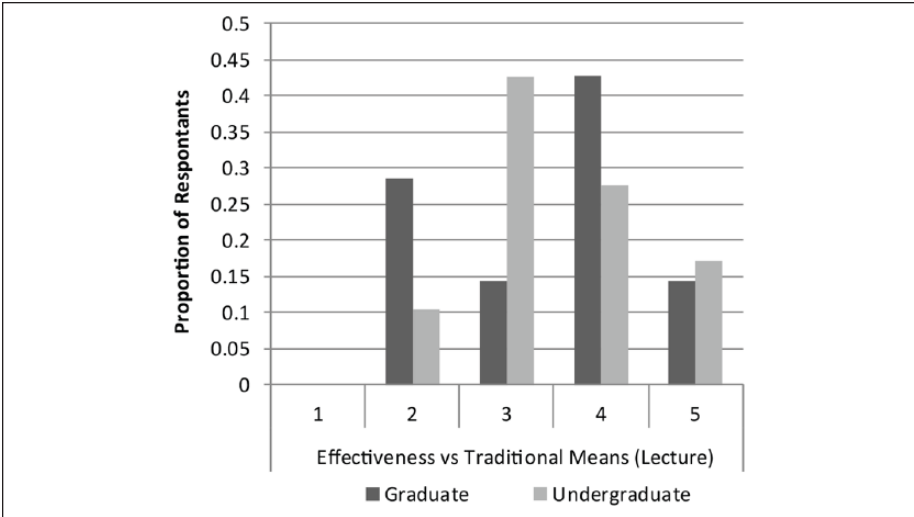


Figure 4. “Would you have understood the history of smuggling and its complexity better if you had spent the class listening to me lecture instead?” (5 No, 3 Similar, 1 Yes).

Lesson 2: Time Constrained Decision-Making May Help Maintain Student Engagement

In 2016, Dr. Sheryl Mills and Ryan Banow served as outside observers for a forty-person undergraduate class (American History Survey Course) playing POLICING THE SOUND. Mills and Banow, noted that only one student did not engage with the game during the first round and by the second round, every student appeared engaged. During the debriefing, Mills and Banow noted that “it seemed that only two or three students did not either ask a question, respond to a question, or offer a point of view.” (Mills, 2016) The high levels of engagement in the debrief after the game well exceeded other activities attempted in that course (primary source readings or short movie clips). The increased student participation during the game did not last beyond the game itself. By the next class, student participation returned to its pre-game levels.

The high levels of participation noted by Mills and Banow in POLICING THE SOUND confirms much of the existing research on games and participation (Nadolny & Halabi, 2016). Student experiences in POLICING THE SOUND, however, also suggests the importance of designing educational activities with time constrained decision making, especially when the progression of all participants is constrained by the slowest group. During each round of POLICING THE SOUND, students had approximately three to four minutes to discuss as a team what goods they were going to ship and whether they were going to smuggle or remain legitimate merchants. The next round did not begin until all teams had chosen. During the first round of the game,

the facilitator occasionally needed to clarify aspects of the rules. This rule explanation extended the timeframe of the first game round unevenly as some students required more clarifications than others did. Disengagement, when it occurred, arose in this first game round after some players completed their turn and found other sources of entertainment while they waited for other groups to catch up. Although other studies have demonstrated the ways that time constrained decisions can reduce performance (Faria, 2001; Gonzalez, 2004), the results in *POLICING THE SOUND* suggest its value in games where the decisions being completed are relatively straightforward and where each group's progress was restricted by the progress of the whole.

Lessons 3: Grouping Players Can Resolve Confusion and Reduce the Time Necessary for Rule Explanation in Analog Games

In digital games, a computer controls the game system and prohibits players from violating the underlying rules of the game. In board and card games, however, players adjudicate their own rule violations eliminating the safeguards found in other media. As a result, a failure to understand the rules in a non-digital game can result in a broken system. While the vast majority of students understood the rules within *POLICING THE SOUND* with relative ease, occasionally one or two students became confused. These misunderstandings did not always become clear until late in the game (or sometimes not until the debriefing) when the students played individually. When players worked as small groups to control one of the game's factions, rule misunderstandings surfaced immediately. Groups asked for clarification within the first round of the game when conflicting understandings of the rules occurred, eliminating the broader impacts of rule misunderstandings. Overall, this study suggests that in classrooms using analog games, arranging students in small groups can help reduce the time necessary for rule explanation, while also minimizing unintended rule violations.

Lesson 4: An Inability to Win Does Not Necessarily Cause Disengagement in Short Games

In *POLICING THE SOUND*, the uneven distribution of goods and capacities makes it difficult if not impossible for some of the communities to win based off the initial victory conditions (total wealth). Across the six times this game has been run in an educational setting, Haida Gwaii and Metlahatla have ended the game with an average wealth of 43, while the customs inspector has averaged 116 and Port Townsend 122. This uneven situation provides an opportunity during the debriefing to discuss the dangers of using aggregate wealth (instead of per capita wealth) to compare communities that ranged in population from a few thousand to tens of thousands (U.S. Census Office, 1892).

While this approach has clear educational payoffs, it also risks creating disengagement among students (Jabbar & Felicia, 2015; Wadhwa & Kim, 2015). In most of the playthroughs, the groups facing significant structural constraints realized their predicament by the middle of the game. Observations of students playing *Poling the*

Sound and their feedback afterwards suggests that, at least in a relatively quick game (~30 minutes), the inability to win does not necessarily compromise student enjoyment, learning, or engagement. Instead, it increased engagement during the debrief afterwards. Students from the losing groups participated heavily in the subsequent discussions about how small shifts in win conditions (aggregate wealth vs. per capita wealth vs. comparing their score to previous groups playing the same team) might have changed the outcome of the game. It also led to fruitful debates about how looking at only a single aspect of an economy (transnational trade) can create a distorted picture of each community's viability and success and how ignoring different cultural perceptions of what success looks like creates misleading depictions of historic communities (Lutz, 2008; Sandage, 2005).

Lesson 5: A Structured Discussion/Debriefing Period is Beneficial Even in Custom Games

The use of POLICING THE SOUND in a wide variety of classroom and non-classroom settings confirms much of the existing literature's conclusions regarding the importance of the discussion or debriefing period after the game (van den Hoogen, Lo, & Meijer, 2016). In the case of POLICING THE SOUND, a ratio of 30 minutes of gameplay time to 40-50 minutes of discussion worked best. This gave students a large amount of time to puzzle through the historic nature of the game and to respond to a wide array of debriefing questions (Appendix A) without resulting in boredom or disengagement. The debrief focused on fostering historical empathy and improving research approaches.

Lesson 6: Custom Built Games Can Teach Research Methodologies by Focusing on Omissions and Errors in the Simulation Itself

The debriefings that followed POLICING THE SOUND suggests that custom-built games and simulations have an advantage over commercial off-the-shelf games because they can model disciplinary approaches, theoretical concepts, and structural limitations in evidence in ways that can open up new avenues for learning (Bird, 2002; Bridge, 2013; Hollander, 1977). The underlying game mechanics in POLICING THE SOUND, for example, are modeled off historic arrest and correspondence records (United States Customs Service. (1886-1912); United States. U.S. District Courts Western District Washington.). This modeling created an unexpected opportunity during the debriefing discussions to push students to interpret how accurate the archival collection (rather than simply the game) represented the past. In this context, POLICING THE SOUND modeled the archival record, for both good and bad, by including those who are caught (the worst smugglers) and all but ignoring those who are not (the best smugglers). In doing so, POLICING THE SOUND presented students with a common research problem - how do you recreate a history where the records are problematic - and asked them to try to solve it using the game as a point of reference. In this sense, customs built games hold particular value for teachers interested in teaching research methodologies

or disciplinary challenges rather than simply content and may be particularly well suited for uses in professional schools and graduate programs.

Lesson 7: Simulating Moral Choices Can Build Historical Empathy by Emphasizing the Structural Factors in Decision Making

Scholars in fields ranging from sociology to nursing have demonstrated the capacity that games, simulations, and roleplaying have for teaching empathy. Games show success conveying empathy when compared to traditional approaches because they allow students to experience structural constraints and social norms first hand (Bearman et al., 2015; Chen et al., 2015; Groves et al., 1996; Hammond, 2006). In history, fostering empathy requires a unique set of approaches as most historians interact only with the fragmented records people leave behind rather than with the people themselves. This interaction with past people at a distance can result in quick judgments, a failure to recognize structural constraints, or a misunderstanding of historical motivations and viewpoints.

The game mechanics in *POLICING THE SOUND* heavily incentivize smuggling through low penalties and high pay offs, but do not necessitate it. Despite the option to remain legitimate, not a single participant or team in any of the classes the game has been run in have played through the game without smuggling at least once. The nature of the setting (players are told this game is about smuggling) and sense of competition creates an environment where smuggling is the norm, rather than an aberration.

When undergraduate students read about smuggling outside of a game setting, they succeeded in identifying the key changes in law and its impact on commercial operations, but tended to miss the actual social environment in which people operated in. The students dismissed smugglers as petty criminals and believed that they would have acted differently if placed in similar circumstances. Ultimately, they failed to identify the ways that individual communities defined what was socially acceptable, which did not always fit neatly with what was legally acceptable.

Undergraduate students came to a different set of conclusions within the game environment. While behavior in real life does not simply mirror behavior in games, as studies on violence and aggression by Gabbiadini et al. (2012), Börsche (2010), Giumetti and Markey (2007), and Möller and Krahé (2009) have shown, simulating illegal behavior in *POLICING THE SOUND* prompted students to reassess the value they placed on social context. During the debriefing, undergraduate students acknowledged that their willingness to break the law was partially encouraged by their peer's behavior and partially the result of the system of reward and punishment simulated by the game. As more and more groups smuggled, smuggling ceased to represent aberrant behavior. Student experiences in *POLICING THE SOUND* suggests that educators interested in encouraging self-reflection or conveying how 'normal behavior' is formulated in society should aim to develop activities that allow students to make difficult moral decisions within a constrained game environment. The act of making meaningful choices in a competitive environment followed by a structured debriefing served as a particularly effective learning tool because it centered student's attention on the wide array of motivations and factors that go into real world decisions.

Conclusion

Games are unlikely to revolutionize the ways that educators and historians teach the past, although they offer a valuable tool for supplementing existing educational strategies. While numerous studies have confirmed that students learn in game-based environments and enjoy the experience, they suggest that the impact of commercial games in classroom settings varies considerably (Ricci et al., 1996; Squire, DeVane, & Durga, 2008; Watson et al., 2011). The kind of game, type of participants (age, gender, and background), and skills being taught all influence uptake (Watson et al., 2011; Yu et al., 2013). As a result, games can be effective at increasing engagement and learning outcomes, but are unlikely to replace monographs, articles, textbooks, lectures, and primary source documents in high school or university classrooms in the near future. Although much research has been done on the impacts of commercially available games - many of which focus heavily around themes of violence, exploration, and conquest - far less research has been done on the use of custom-made games that link course content and gameplay. The high learning curve and time investment required to develop these kinds of interactive activities and relative disparity in their resulting quality partially explains this trend.

Participant feedback reports, facilitator observations, and external observations regarding *POLICING THE SOUND* suggests the viability of custom-made games for teaching not only historical content, but also research methodologies and historic empathy. Presenting these ideas through a game rather than through lectures or seminar discussions encouraged a broader array of student engagement and resulted in students adopting a more reflective stance on the importance of structural factors in influencing perceptions of right and wrong. Outside of the historical profession, the pilot run of *POLICING THE SOUND* suggests several broad conclusions for teachers interested in developing custom-built educational board games. In particular, the results from the pilot run of *POLICING THE SOUND* emphasized that groups of players can resolve confusion more efficiently than they can as individuals, that time constrained decision-making may help maintain student engagement, and that an inability to win does not necessarily cause disengagement in shorter games. Finally, the pilot helped to confirm the importance of a structured debrief for attaining educational goals even in custom-built games that closely model the exact processes being taught.

Appendix A

Debriefing Questions

1. Did the game reflect the initial backstory? Why or why not?
2. Who are the Indigenous communities in this game? How do you know? How does this game represent smuggling? Does the game represent Europeans differently than First Nations?
3. How does geography play into the kinds of goods each community had access to and the kinds of goods that players smuggled?
4. Did every community feel like they had a chance to win? How would the game change if the win condition shifted from most total money to most per capita?

How does setting wealth as the victory condition shape the way this history is represented?

5. Did this game make historical arguments? If so what were they? Did any of the communities choose not to smuggle the entire game? Given the option, why did all of you break the law? What are the limitations of building a game/set of arguments around customs inspector's records?
6. What communities did the customs inspectors look at? Why? Was this an effective strategy? Did this reflect historical practices?
7. If you were asked in a term paper to judge the historic merit of a game, what questions might you ask? Are these questions similar or different to what you would ask of another source?

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